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CSC 1310

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Program 3 Proposal

# Project Proposal

***Partner Identification***

For this project, I plan to work alone in implementing each feature.

***Goals of Completed Project***

The product I am developing is a health tracker—primarily for foods and their nutritional value. The user can search through a database of foods and enter in the amount they plan to eat, where the program will then log it and track all the macronutrients the food from the database provides.

The tracker allows the user to add food to the database and to edit the macronutrients from the custom food they provide. They will not be able to edit the foods information from the universal database in food.csv, but they can upload and customize their own foods from custom.csv.

The data will be stored based on user and day. When the logged-in user enters the day, they can enter in the foods they have eaten, and the program will tally up the individual macros and calories for that day. If the user hits their personal macro/calorie goal, they will be congratulated.

***Data Structures Implemented***

The data structures used will be Hash tables and 2D arrays. The hash table will be used to store and look up food information based on the key provided by the user to log into. From there, the table will pull an array with the food's information and use each index from the array to track the specific nutrient.

The 2D array will store the user's input data for their weight, calorie limit, calorie intake for the day, and a Boolean for if they exceeded their calorie limit. The rows will be divided by date and the columns by measurement.

***Three Planned Application Features***

***1.)*** Log food Item from food.csv

- ***a.*** Allow user to enter in food name and search for any potential matches in the universal food database
- ***b.*** Allow user to create custom food with custom macros and calories
- ***c.*** Allow user to delete/edit custom food but NOT food from universal list

***2.)*** Display User information

- ***a.*** User may choose to see their current caloric/macro intake for the day compared to their goal/limit
- ***b.*** User may edit weight, calorie limit, or macro goals

***3.)*** Sort through dates with logged food

- ***a.*** Allows user to review and edit their progress from a previous date

***Detailed Tasks***

| Detailed Task Description | Due Date | Current Status |
|---|---|---|
| .csv files for food and custom foods | 11/12/26 | COMPLETE |
| Skeleton for Food Class | 11/15/26 | INCOMPLETE |
| Skeleton for UserInfo Class | 11/15/26 | INCOMPLETE |
| menu and math functions | 11/18/26 | INCOMPLETE |
| Implement .csv loading | 11/20/26 | INCOMPLETE |
| sort algorithms for food.csv and custom.csv | 11/23/26 | INCOMPLETE |
| Implement 2D array and array logic for user info (includes name, date logged, weight, calorie limit, macro goals) | 11/25/26 | INCOMPLETE |
| deletion and insertion algorithm for custom.csv | 11/25/26 | INCOMPLETE |
| Goal congratulation trigger | 11/27/26 | INCOMPLETE |
| Debugging and testing | 12/1/26 | INCOMPLETE |